

COM 3240

REINFORCEMENT LEARNING



DALL·E

IMMEDIATE REWARDS

SLOT MACHINES: ONE-ARM BANDITS



Charles Fey



End of 19th century

DALL·E

IMMEDIATE REWARDS

HOUSE EDGE

$$E[R] = \sum_i P(R = r_i) r_i$$

Negative rewards: bet

Positive rewards: payout

$$E[R] = P_{win}(\mathbf{Payout}) + (1 - P_{win})(-\mathbf{Bet})$$

$$\mathbf{House\ Edge\ (\%)} = (-1) \left(\frac{E[R]}{\mathbf{Bet}} \right) 100 \%$$

IMMEDIATE REWARDS

HOUSE EDGE

probability of winning is $1/37$

bet £1

$$E[R] = \left(\frac{1}{37}\right) £35 + \left(1 - \frac{1}{37}\right)(-£1) = -£0.027$$

payout of £35

$$\text{House Edge} = (-1) \frac{-£0.027}{£1} = 2.7 \%$$

IMMEDIATE REWARDS

BANDITS

Reward distributions are unknown

$$q^*(a) \doteq E[R_t | A_t = a]$$

In RL we obtain an estimate via sampling

Q-value : estimate of true expected reward



DALL·E

OPTIMISATION

GRADIENT DESCENT

Desirable: A system (an agent) that observes rewards and estimates expectations

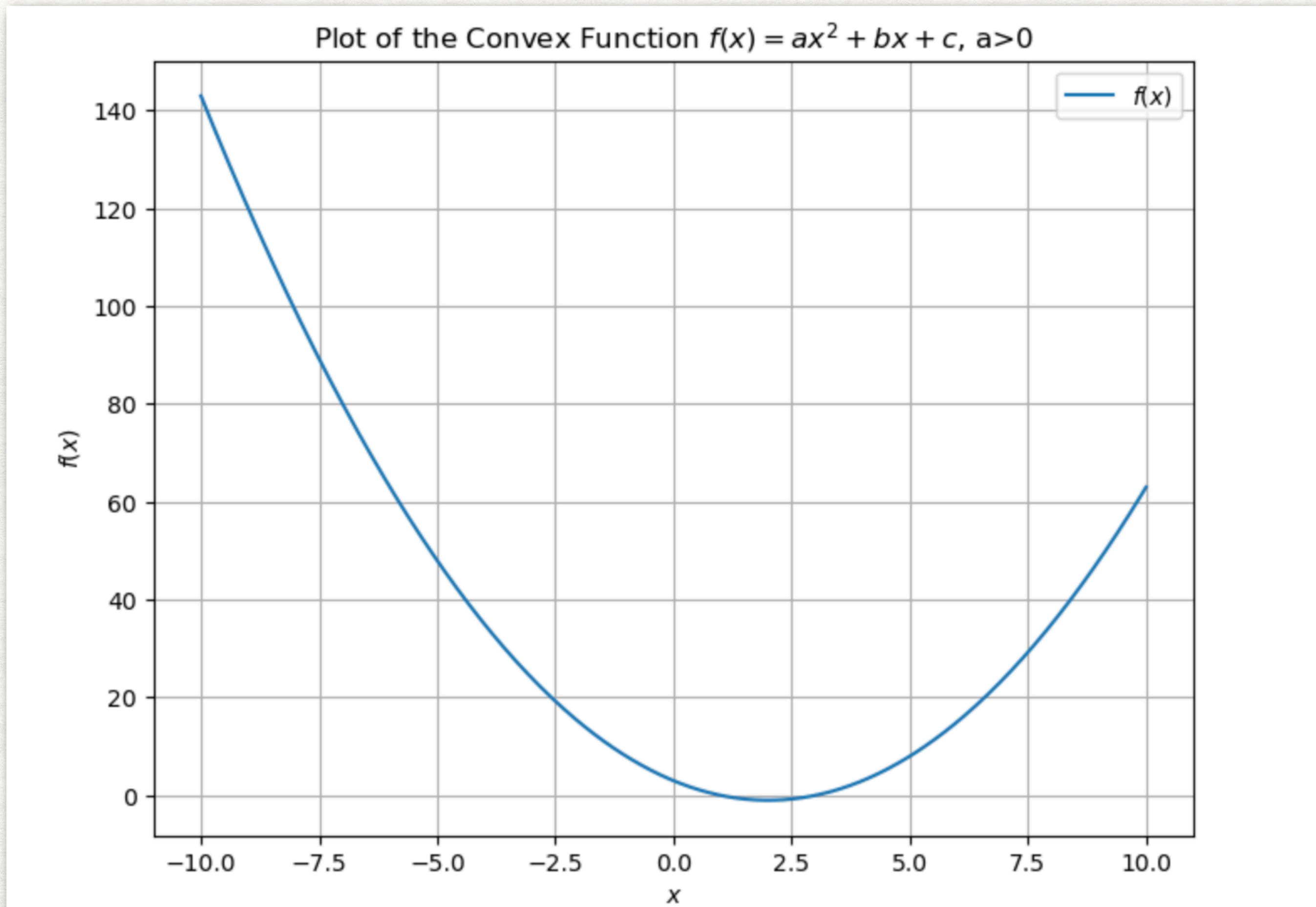
The system has parameters that we need to select appropriately (optimise)

$f(x_1, x_2, \dots, x_n)$
parameters

f System's performance: maximise
System's error : minimise

OPTIMISATION

GRADIENT DESCENT

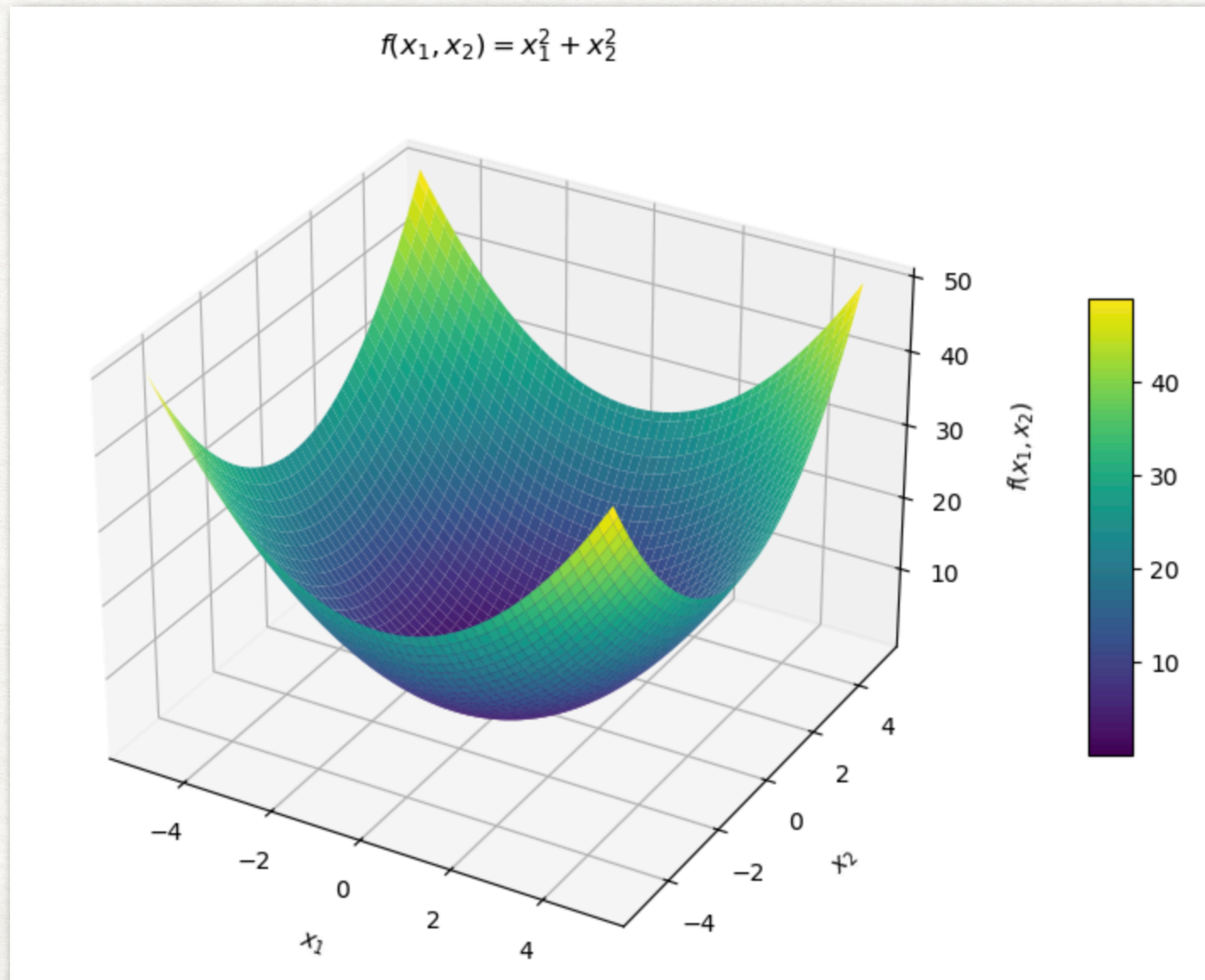


$$\frac{d}{dx}(ax^2 + bx + c) = 2ax + b$$

$$2ax + b = 0$$

OPTIMISATION

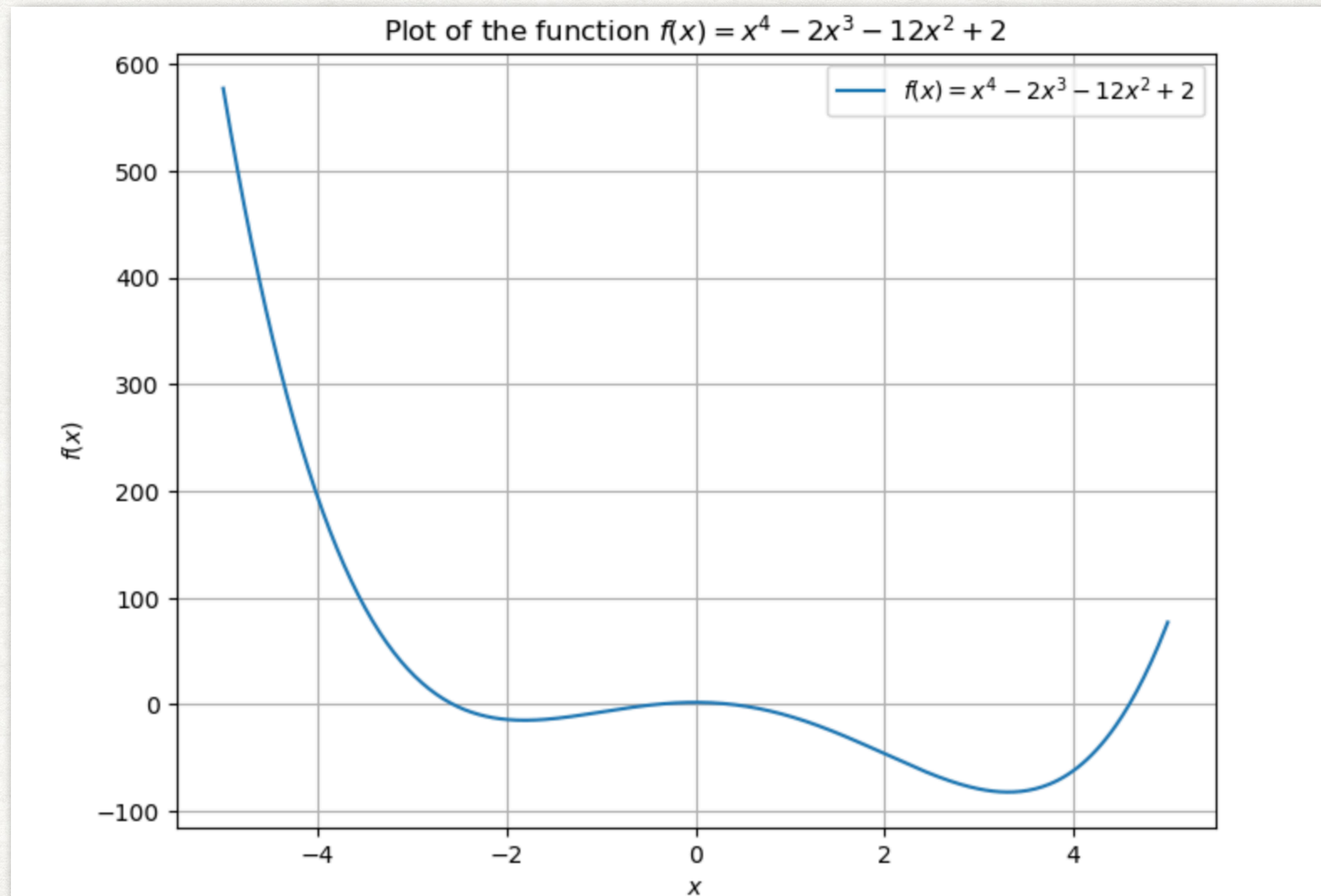
GRADIENT DESCENT



$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right] = 0$$

OPTIMISATION

GRADIENT DESCENT



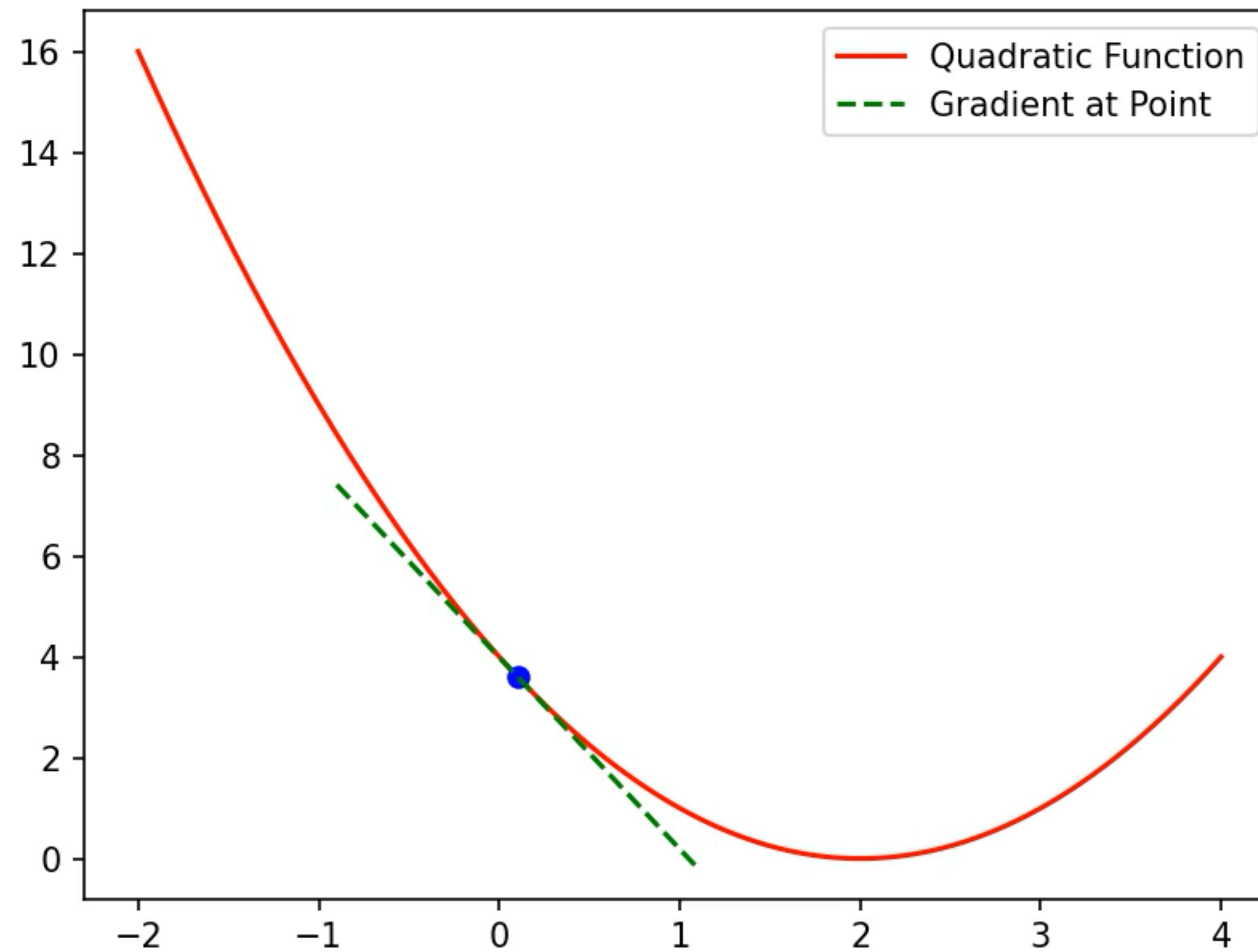
$$\mathbf{x}_{i+1} = \mathbf{x}_i - \eta \nabla f(\mathbf{x}_i)$$

$$\Delta \mathbf{x} = -\eta \nabla f(\mathbf{x})$$

$$\Delta \mathbf{x} = \mathbf{x}_{i+i} - \mathbf{x}_i$$

OPTIMISATION

GRADIENT DESCENT



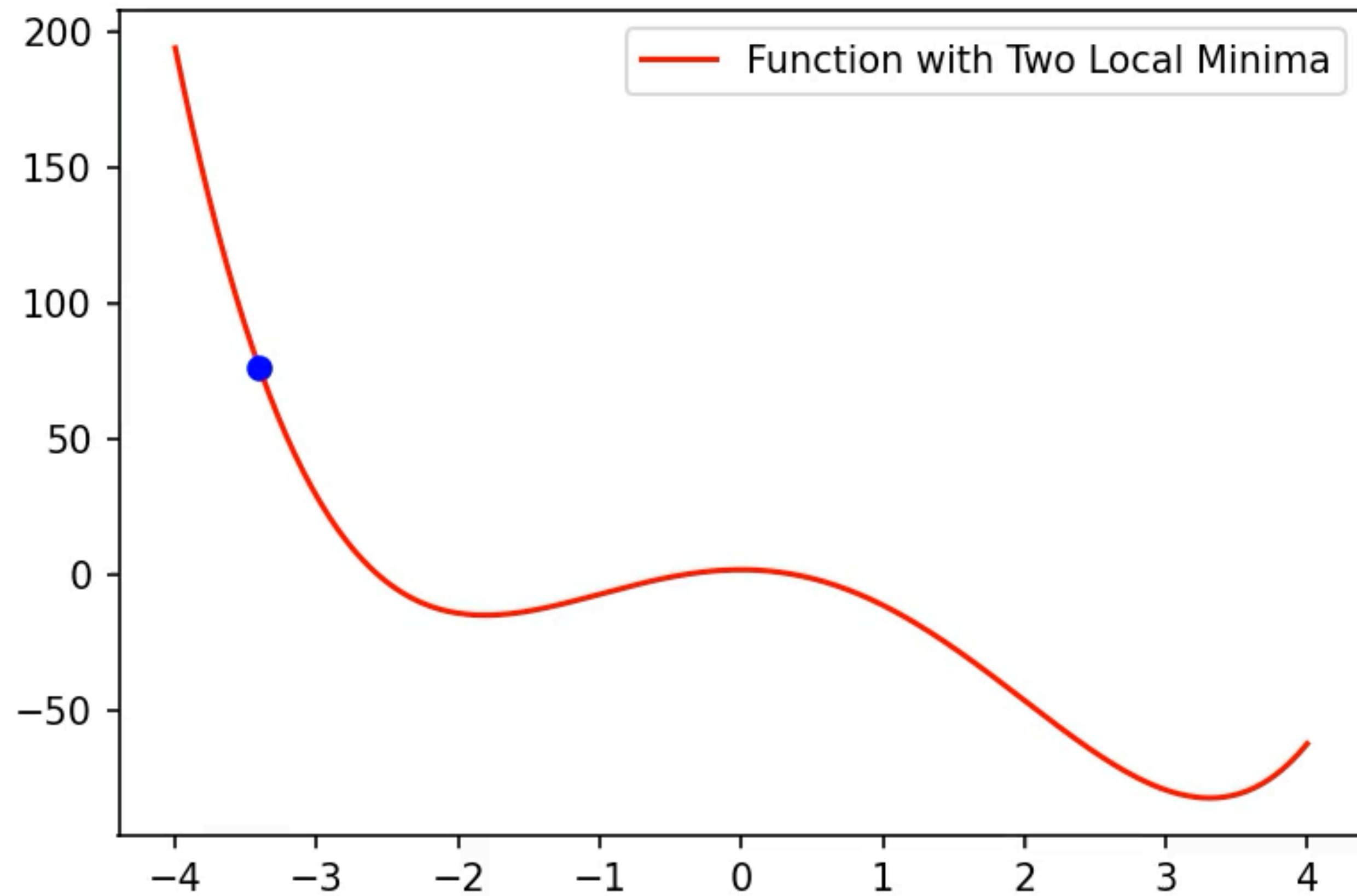
$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\Delta \mathbf{x} = -\eta \nabla f(\mathbf{x})$$

$$\Delta \mathbf{x} = \mathbf{x}_{i+1} - \mathbf{x}_i$$

OPTIMISATION

GRADIENT DESCENT



IMMEDIATE REWARDS

BANDITS

$$E[R] = \sum_i r_i P(R = r_i) \quad \text{Unknowns: } r_i, P(R = r_i)$$

$$Q = \frac{1}{n} \sum_{i=1}^n r_i \quad \text{Stationary}$$

$$\text{Error}_b = \frac{1}{2} \sum_i (r_i - Q)^2 \quad \text{Minimise}$$



DALL·E

IMMEDIATE REWARDS

BANDITS

$$\mathbf{Error}_b = \frac{1}{2} \sum_i^n (r_i - Q)^2 \quad \text{Minimise}$$

$$\Delta Q = -\eta \frac{\partial \mathbf{Error}_b}{\partial Q} = \eta \sum_{i=1}^n (r_i - Q)$$

The result is the same as: $Q = \frac{1}{n} \sum_{i=1}^n r_i$

IMMEDIATE REWARDS

BANDITS

$$\mathbf{Error}_b = \frac{1}{2} \sum_i^n (r_i - Q)^2 \quad \text{Batch}$$

$$\Delta Q = -\eta \frac{\partial \mathbf{Error}}{\partial Q}$$

$$\mathbf{Error}_o = \frac{1}{2} (r - Q)^2 \quad \text{Online}$$

Non-stationary

Mini-batch

IMMEDIATE REWARDS

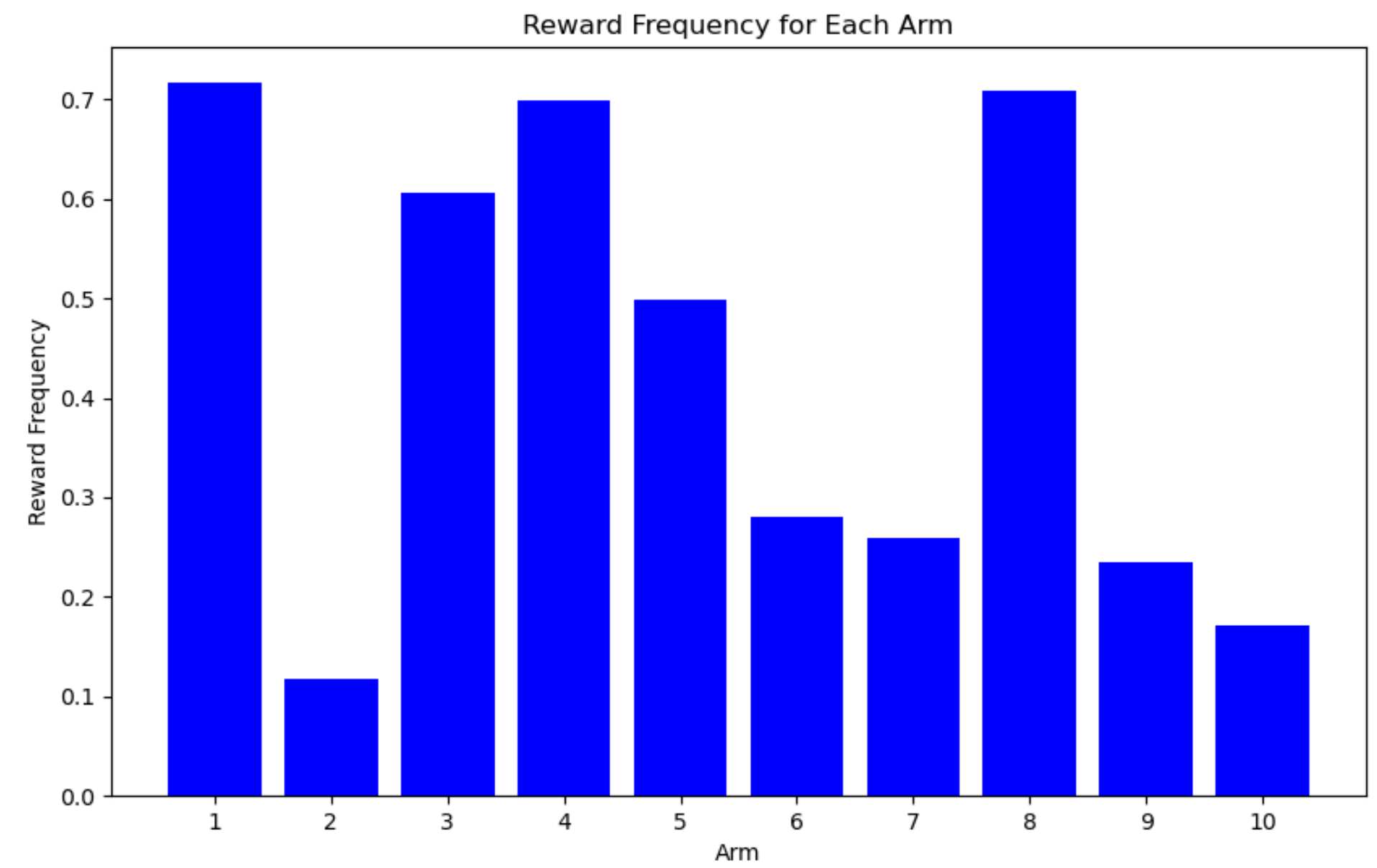
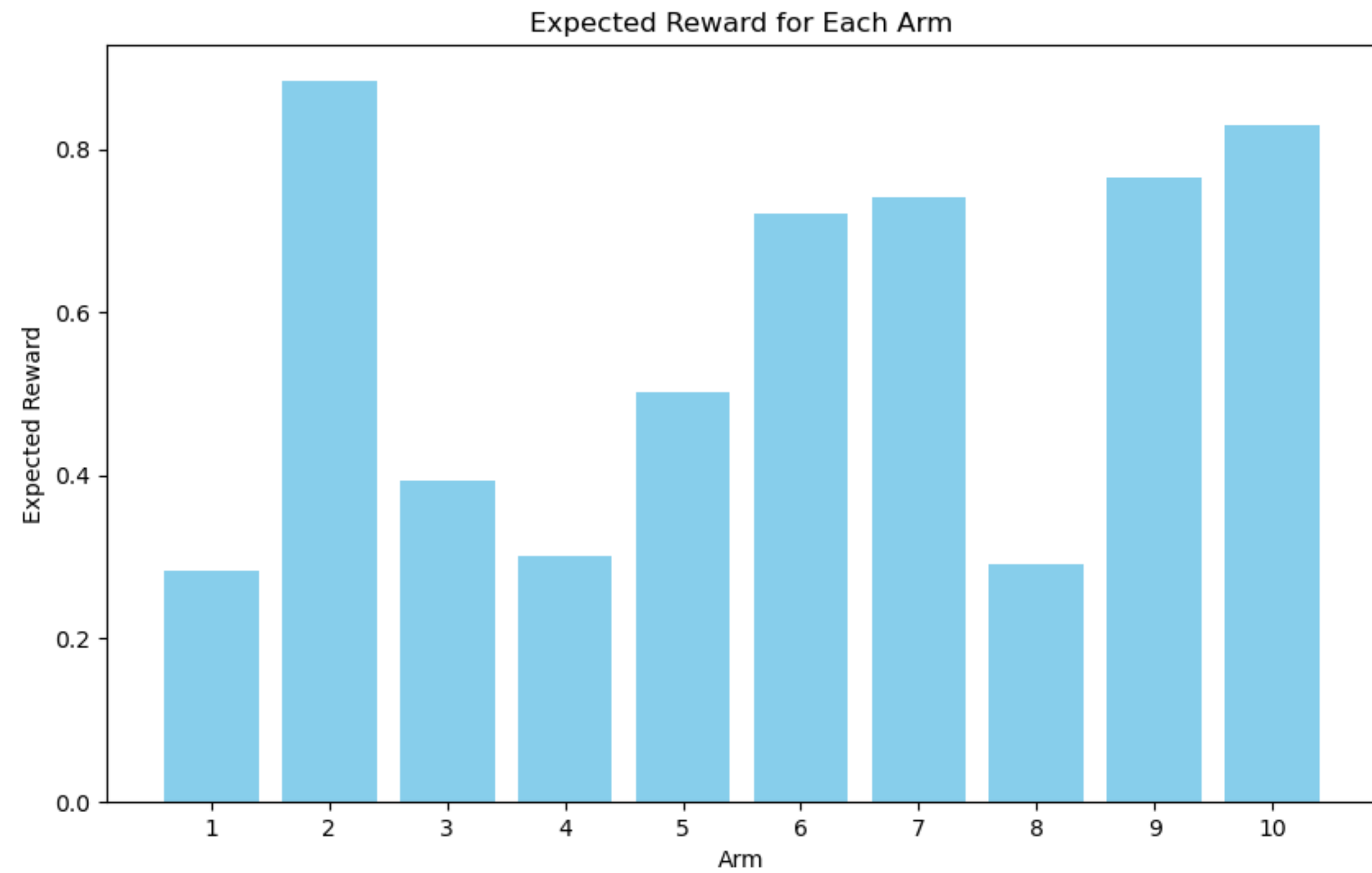
BANDITS



DALL·E

IMMEDIATE REWARDS

BANDITS



POLICIES

THE EXPLORATION EXPLOITATION DILEMMA



DALL·E

POLICIES

THE EXPLORATION EXPLOITATION DILEMMA

Greedy $a_t = \operatorname{argmax}_a Q_t(a)$

Optimistic Greedy: initialise Q-values unrealistically high

Epsilon-Greedy : explore with probability epsilon, greedy otherwise

Softmax: $P(a) = \frac{e^{Q_t(a)/\tau}}{\sum_b e^{Q_t(b)/\tau}}$

IMMEDIATE REWARDS

BANDITS

- Initialise Q-values, and count N
- For each time step
 - Select an action using policy
 - Take action and observe reward
 - Increase the count N
 - Update estimates $Q(A_{t+1}) = Q(A_t) + (r_t - Q(A_t))/N(A_t)$

IMMEDIATE REWARDS

BANDITS

- Initialise Q-values, and count N
- For each time step
 - Select an action using policy
 - Take action and observe reward
 - Increase the count N
 - Update estimates $Q(A_{t+1}) = Q(A_t) + \eta(r_t - Q(A_t))$

IMMEDIATE REWARDS

BANDITS

$$\mathbf{Error}_b = \frac{1}{2} \sum_a \sum_i (r_i - Q(a))^2 \mathbf{1}_{A_i=a}$$

$$\mathbf{1}_{A_i=a} = \begin{cases} 1 & \text{if } A_i = a \\ 0 & \text{otherwise.} \end{cases}$$

IMMEDIATE REWARDS

BANDITS

$$\Delta Q(a) = -\eta \frac{\partial \mathbf{Error}_b}{\partial Q(a)} = \eta \sum_i (r_i - Q(a)) \mathbf{1}_{A_i=a}$$

THANK YOU!